

these peripherals are not used.

- $V_{DDSMPS} = 2\text{ V to }3.6\text{ V}$

V_{DDSMPS} is the external power supply for the SMPS step down converter. It is provided externally through V_{DDSMPS} supply pin, and shall be connected to the same supply as V_{DD} .

- V_{LXSMPS} is the switched SMPS step down converter output.
- V_{15SMPS} are the power supply for the system regulator. It is provided externally through the SMPS step down converter V_{LXSMPS} output.

Note: The SMPS power supply pins are available only on a specific package with SMPS step down converter option.

- $V_{DD12} = 1.05\text{ to }1.32\text{ V}$

V_{DD12} is the external power supply bypassing the internal regulator when connected to an external SMPS. It is provided externally through V_{DD12} pins and only available on packages with the external SMPS supply option. V_{DD12} does not require any external decoupling capacitance and cannot support any external load.

Note: The V_{DD12} power supply pins are available only on a specific package with external SMPS option.

- $V_{DDUSB} = 3.0\text{ V to }3.6\text{ V}$

V_{DDUSB} is the external independent power supply for USB transceivers. The V_{DDUSB} voltage level is independent from the V_{DD} voltage. V_{DDUSB} should be preferably connected to V_{DD} when the USB is not used.

The V_{DDUSB} power supply may not be present as a dedicated pin, but to be internally bonded to V_{DD} . For such devices, V_{DD} has to respect the V_{DDUSB} supply range when the USB is used.

- $V_{DDIO2} = 1.08\text{ V to }3.6\text{ V}$

V_{DDIO2} is the external power supply for 14 I/Os (Port G[15:2]). The V_{DDIO2} voltage level is independent from the V_{DD} voltage and should preferably be connected to V_{DD} when PG[15:2] are not used.

- $V_{BAT} = 1.55\text{ V to }3.6\text{ V}$

V_{BAT} is the power supply for RTC, external clock 32 kHz oscillator and backup registers (through power switch) when V_{DD} is not present. V_{BAT} is internally bonded to V_{DD} for small packages without dedicated pin.

- V_{REF-}, V_{REF+}

V_{REF+} is the input reference voltage for ADCs and DACs. It is also the output of the internal voltage reference buffer when enabled.

When $V_{DDA} < 2\text{ V}$, V_{REF+} must be equal to V_{DDA} .

When $V_{DDA} > 2\text{ V}$, V_{REF+} must be between 2 V and V_{DDA} .

V_{REF+} can be grounded when ADC and DAC are not active.

The internal voltage reference buffer supports two output voltages, which are configured with VRS bit in the VREFBUF_CSR register:

- V_{REF+} around 2.048 V. This requires V_{DDA} equal to or higher than 2.4 V.
- V_{REF+} around 2.5 V. This requires V_{DDA} equal to or higher than 2.8 V.

On some packages, V_{REF-} and V_{REF+} pins are not available. When not available on the package, they are internally bonded to respectively V_{SSA} and V_{DDA} .